



LogiChain

An end-to-end platform connecting products, warehouses, and customers through seamless digital operations

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Addressing Modern Supply Chain Challenges



The Problem

Traditional supply chain systems struggle with fragmented workflows, lack of real-time visibility, and poor communication between stakeholders. Businesses face inventory inaccuracy, delayed shipments, and inefficient return processing.



Automated Workflows

Streamlined order processing from creation to delivery



Multi-Warehouse

Centralized management of distributed inventory locations



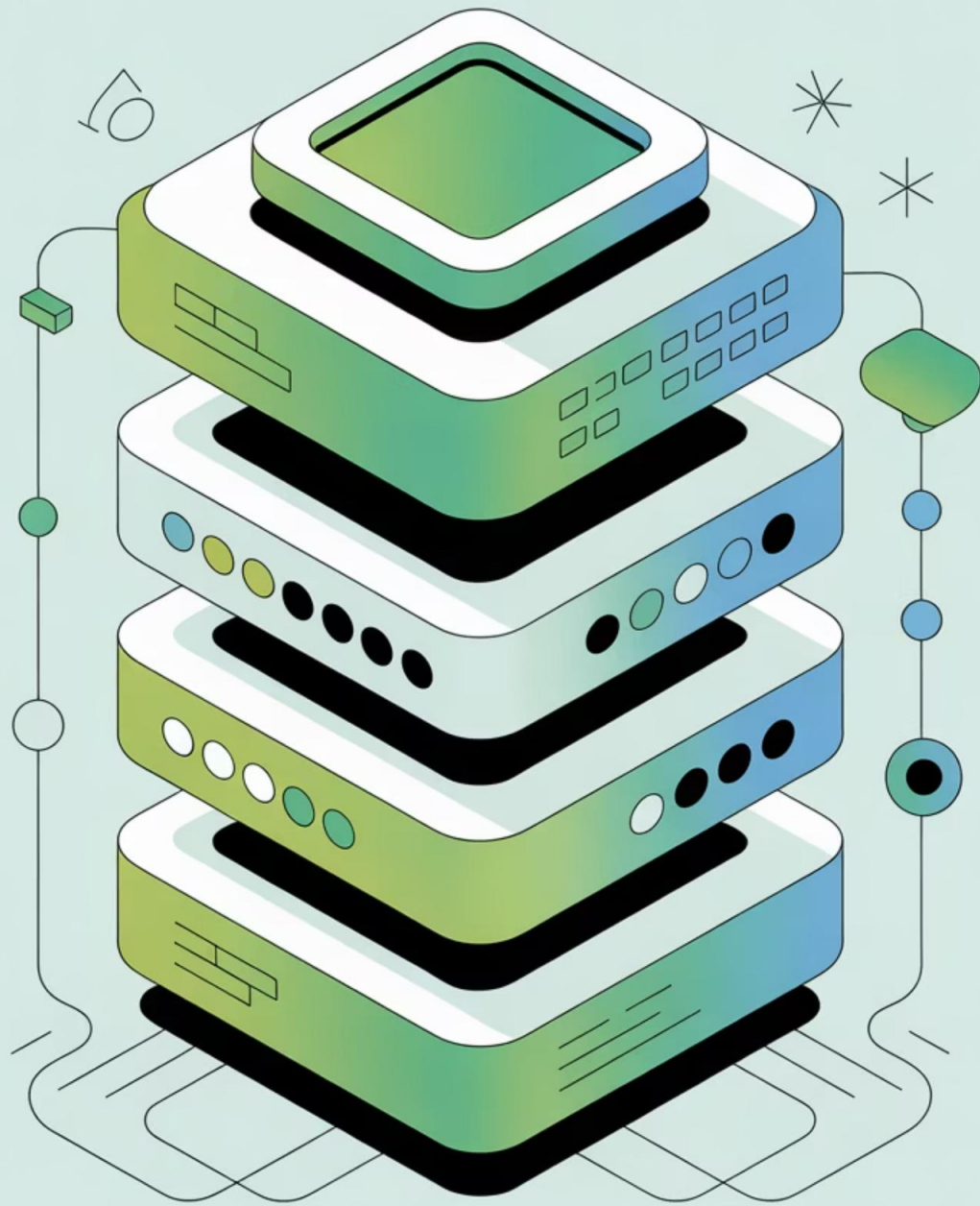
Real-Time Sync

Instant inventory and order status updates across all modules



Complete Visibility

End-to-end shipment tracking with real-time events



Robust Architecture Built on Modern Technologies

React Frontend



Responsive UI with React Router, Axios API calls, TailwindCSS styling, and React Toastify notifications

Spring Boot Backend



Business logic, JWT authentication, role-based security, RESTful APIs, and ModelMapper conversions

MySQL Database



Data integrity with encrypted passwords, stateless JWT tokens, and comprehensive validation

Enterprise-grade security includes BCrypt password hashing, CORS configuration, CSRF protection, and HTTP-only secure cookies for token storage.

Role-Based Access Control System

Five distinct user types with specific permissions ensuring data security and operational efficiency



Admin

Complete system oversight, user approval, product management, warehouse monitoring, and analytics configuration



Product Manager

Creates and manages product catalog with full CRUD operations, controls inventory stock levels, handles restocking



Warehouse Manager

Manages physical inventory, processes orders through lifecycle, creates shipments with real-time tracking



Customer Support

Reviews return requests, updates order statuses, manages queries, tracks shipments, monitors notifications



Customer

Browses products, places orders, tracks shipments, requests returns, and manages personal profile

Smart Product & Inventory Control

Product Management

Product Managers create products with SKU, name, category, price, weight, description, and images. The system supports categorization with search and filter capabilities. Unique Product Manager ownership ensures each manager only sees and controls their products through createdBy field filtering.

Stock Tracking

- **Total Stock:** Sum of available and reserved units
- **Available Stock:** Ready for immediate sale
- **Reserved Stock:** Allocated to pending orders

Real-Time Synchronization

Stock updates instantly across all modules, preventing overselling through available stock checks

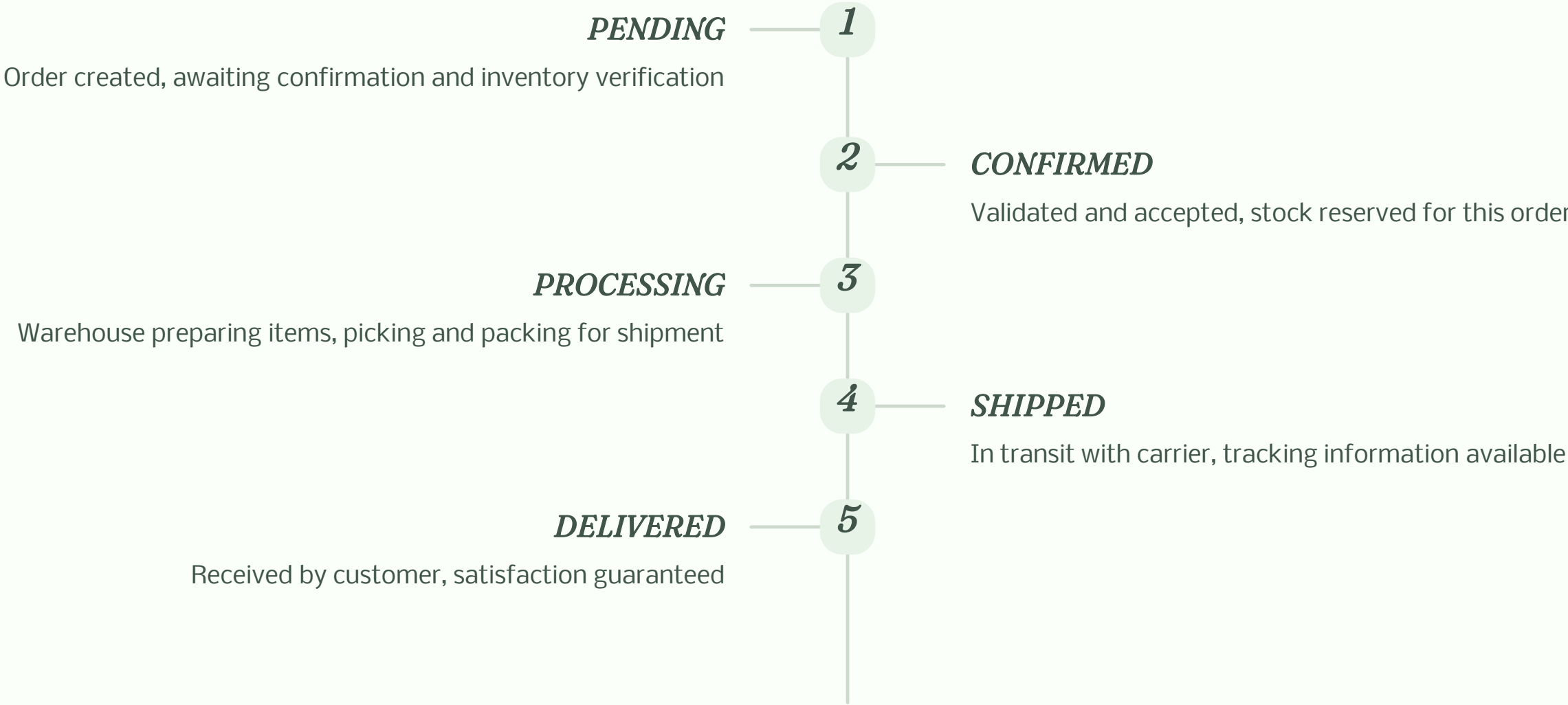
Multi-Warehouse Distribution

Same product across different warehouses with independent stock tracking and automatic aggregation

Low Stock Alerts

Dashboard displays products with less than 50 units remaining, triggering restock requests

Streamlined Order Processing Pipeline



Automatic Stock Reservation

System reserves required inventory when order is confirmed, preventing overselling

Unique Order Numbers

ORD-YYYY-NNNN format with automatic status tracking and payment integration

Real-Time Notifications

Customers informed at each status change through email, SMS, and in-app alerts

Real-Time Visibility Across the Supply Chain

Shipment Lifecycle

01

CREATED

Shipment initiated with carrier selection

02

PICKED_UP

Carrier collected package from warehouse

03

IN_TRANSIT

En route to destination

04

OUT_FOR_DELIVERY

Final mile delivery in progress

05

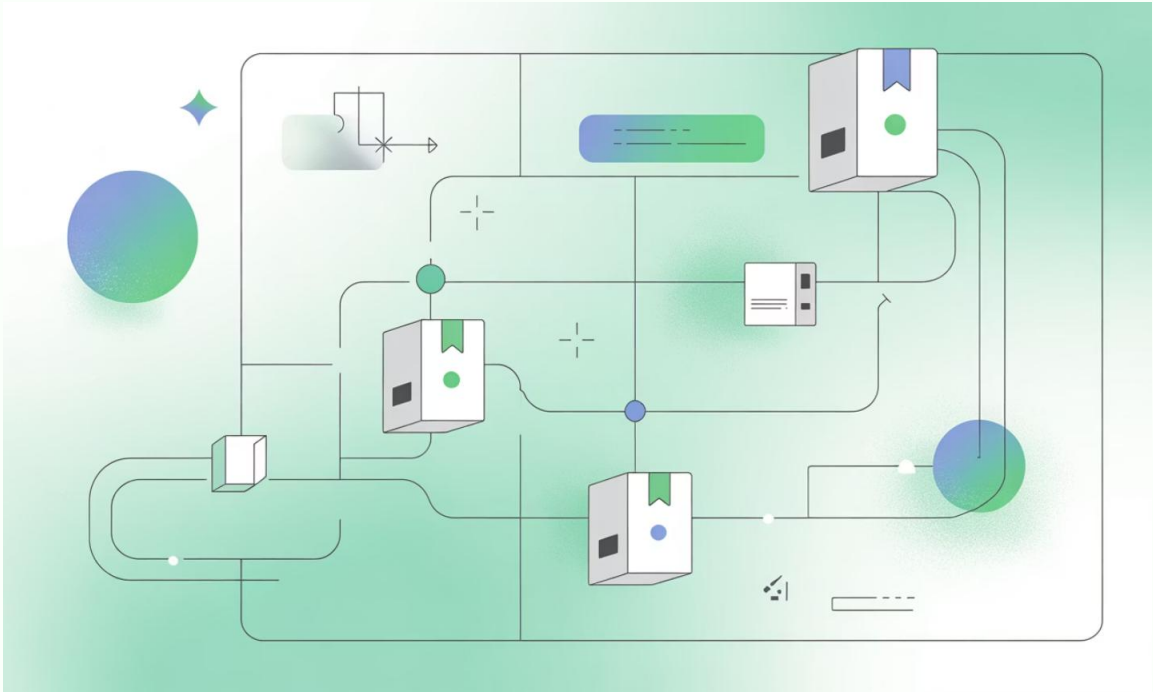
DELIVERED

Successfully received



Multiple Carriers

FedEx, UPS, DHL, USPS, and more supported



Complete Audit Trail

Each tracking event captures timestamp, location, and description. Customers view detailed timeline with visual progress indicators and estimated delivery updates.



Unique Tracking

Auto-generated numbers for every shipment

Complete Return Processing Workflow



Customer Experience

Customers request returns for delivered orders by selecting reason: Defective, Wrong Item, Not as Described, Changed Mind, or Other. System auto-generates return numbers (RET-timestamp-orderID format) and calculates refund amount from original order total.

- Minimum 10-character description required
- Dynamic status badges show next steps
- Complete return history across all pages



Return Tracking
Full visibility into request status

Enterprise-Grade Security Implementation



JWT Authentication

Stateless sessions with secure token-based API access and automatic expiration



BCryptPasswordEncoder

Secure password hashing with salt rounds and token validation on every request



Spring Security

Role-based access control at method level with @PreAuthorize annotations

Registration & Approval

- Email verification required
- Password strength enforcement
- Role-based admin approval for Product Managers, Warehouse Managers, Customer Support
- Auto-approval for Customers

API Security

- CORS policy for React frontend
- CSRF protection for state changes
- HTTP-only secure cookies
- 401 Unauthorized or 403 Forbidden responses



Project Impact & Roadmap

60%

Time Reduction

Faster order processing vs traditional systems

99.5%

Inventory Accuracy

Real-time stock synchronization

5

User Roles

Complete role-based access control

100%

Visibility

End-to-end supply chain tracking

Future Enhancements

Mobile Applications

iOS and Android apps for on-the-go management

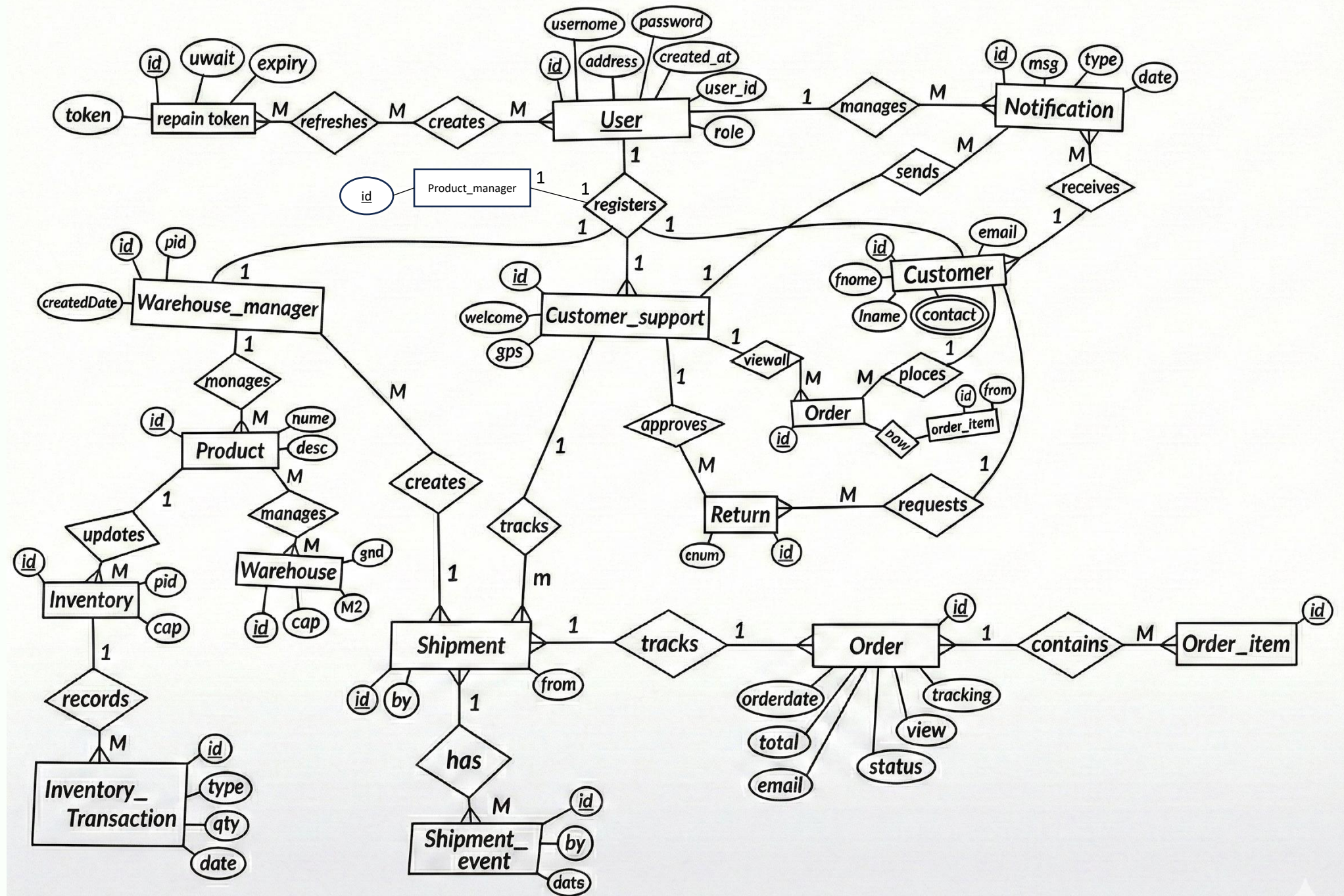
E-Commerce Integration

Connect with Shopify, WooCommerce platforms

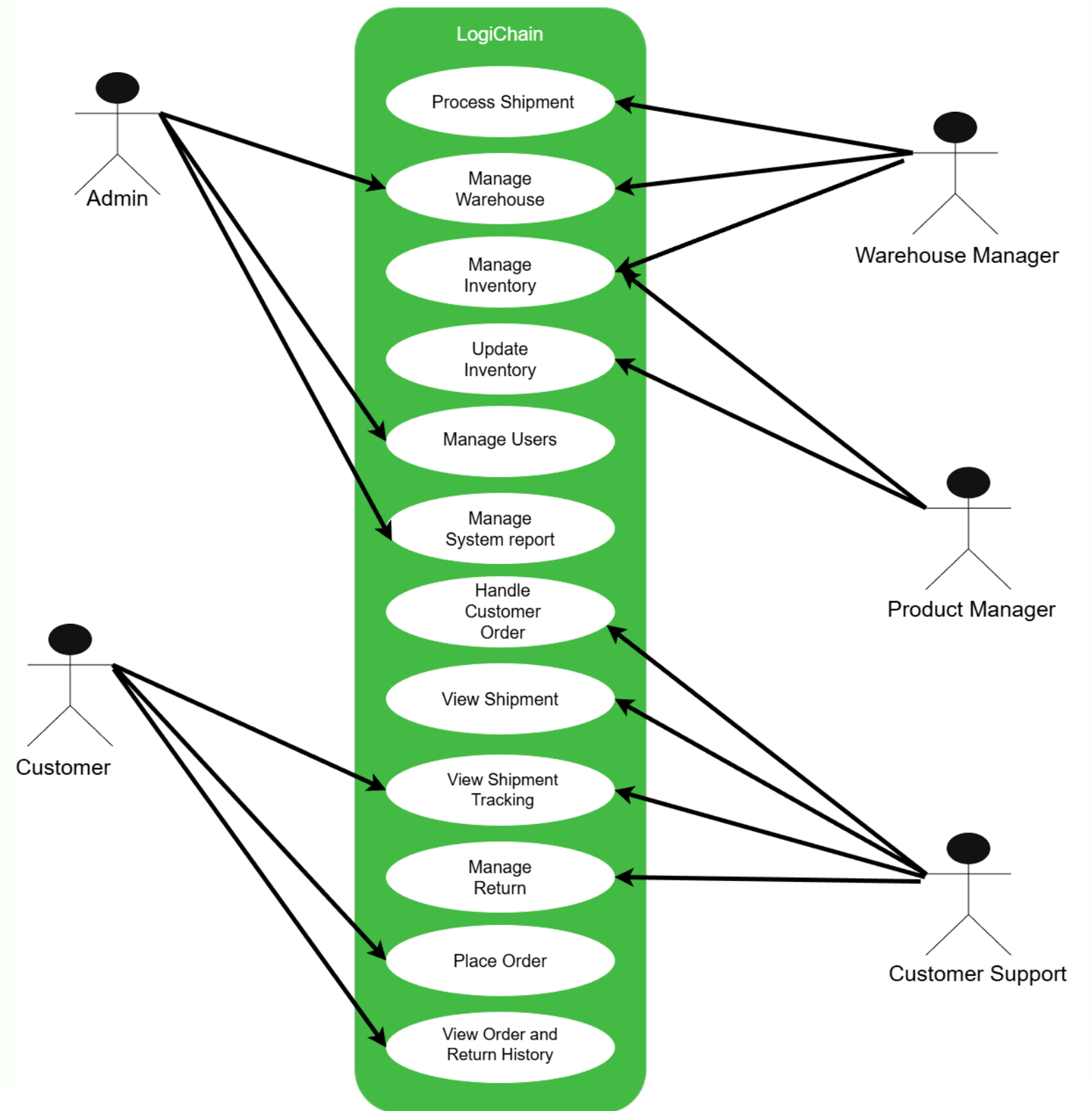
LogiChain demonstrates modern full-stack development practices, enterprise-level architecture, comprehensive security, and user-centric design—making it a robust foundation for supply chain digitization.



ER-Diagram



Use Case Diagram



Real-Time Shipment Tracking & Inventory Management

Complete visibility across your entire supply chain. Track shipments, manage inventory, and streamline warehouse operations in one powerful platform.

[Start Free Trial](#)[Sign In](#)

Landing Page

Key Features



Real-Time Tracking

Track shipments from warehouse to customer with live updates and location tracking.



Inventory Management

Manage stock levels, warehouse operations, and get low-stock alerts automatically.



Analytics & Reports

Get insights into orders, shipments, and inventory with powerful analytics.

Auth Pages



LogiChain

Supply Chain Management

Welcome Back

Sign in to continue to your account

Email Address

you@example.com

Password

.....

[Forgot Password?](#)

Sign In

Don't have an account? [Create Account](#)



Create Account

Join LogiChain today

Username

johndoe

Email Address

you@example.com

Password

.....

Confirm Password

.....

Select Your Role



Customer

Browse and order products



Warehouse Manager

Manage inventory and warehouses



Customer Support

Handle customer inquiries



Product Manager

Manage product catalog and pricing

Create Account

Already have an account? [Sign In](#)

Admin Dashboard

LogiChain

Dashboard

Users

Products

Warehouses

Orders

Inventory

Carriers

Analytics

Settings

Profile

Logout

Welcome back, admin1!

Admin Dashboard

Overview of entire system



12

Total Users



14

Total Orders



16

Products



4

Warehouses

Recent Orders

ORD-ML48U23F-3GTNJJ
\$63.00

DELIVERED

ORD-ML47U6XM-B252PJ
\$21.00

DELIVERED

ORD-ML47TJEP-ZQL9LM
\$89.99

SHIPPED

System Overview

Active Users

11

Pending Orders

4

Low Stock Items

0

Active Shipments

0

Warehouse Dashboard

LogiChain

Welcome back, ware!

Dashboard

Inventory

Warehouses

Orders

Shipments

Products

Profile

Logout

Warehouse Dashboard

Manage inventory and shipments



4726

Total Items



0

Orders Today



3

Shipments



3

Low Stock

ORD-2024-0002
john_doe

Ship Now

ORD-2024-0003
jane_smith

Ship Now

ORD-2024-0004
jane_smith

Ship Now

Low Stock Alerts

Product
30 units left

Restock

Product
40 units left

Restock

Product
30 units left

Restock

Product Dashboard

 LogiChain

 Dashboard

 Products

 Profile

 Logout

localhost:5173/products

✓ Welcome back, pmanager!

Product Manager Dashboard

Manage your product catalog and pricing

Total Products

16



Active Products

16



Catalog Value

\$6000.85



Low Stock Items

0



Recent Products

SKU	NAME	PRICE	WEIGHT
ELEC-PHN-002	qwe	\$21.00	12 kg
OTHER-BACKPACK-001	Travel Backpack 40L	\$89.99	1.2 kg
OTHER-WATERBOTTLE-001	Insulated Water Bottle	\$34.99	0.4 kg
FURN-CHAIR-001	Ergonomic Office Chair	\$399.99	15 kg
FURN-DESK-001	Standing Desk Adjustable	\$599.99	35 kg

Quick Actions

Manage Products

Add New Product

Customer Support Dashboard

LogiChain

Dashboard

All Orders

Returns

Customers

Tracking

Notifications

Profile

Logout

Welcome back, csupport!

Support Dashboard

Manage customer orders and returns



0

Orders Today



0

Return Requests



5

Customers



0

Urgent Issues

Recent Support Tickets

ORD-2024-0001

PENDING

Customer: john.doe@email.com

Amount: \$2499.99

View

Contact

Return Requests

No pending return requests

Customer Dashboard

 LogiChain

 Dashboard

+ New Order

 My Orders

 Returns

 Profile

 Logout

✓ Welcome back, cust!

My Dashboard

Track your orders, shipments, and returns



3

Total Orders



1

Active Shipments



0

Pending Returns



2

Delivered

Recent Orders

[View All →](#)



ORD-ML48U23F-3GTNJJ
\$63.00

DELIVERED

[Track](#)

[Details](#)



ORD-ML47U6XM-B252PJ
\$21.00

DELIVERED

[Track](#)

[Details](#)



ORD-ML47TJEP-ZQL9LM
\$89.99

SHIPPED

[Track](#)

[Details](#)



New Order
Place a new order



Track Orders
Track your shipments



Returns
Manage return requests



THANK YOU